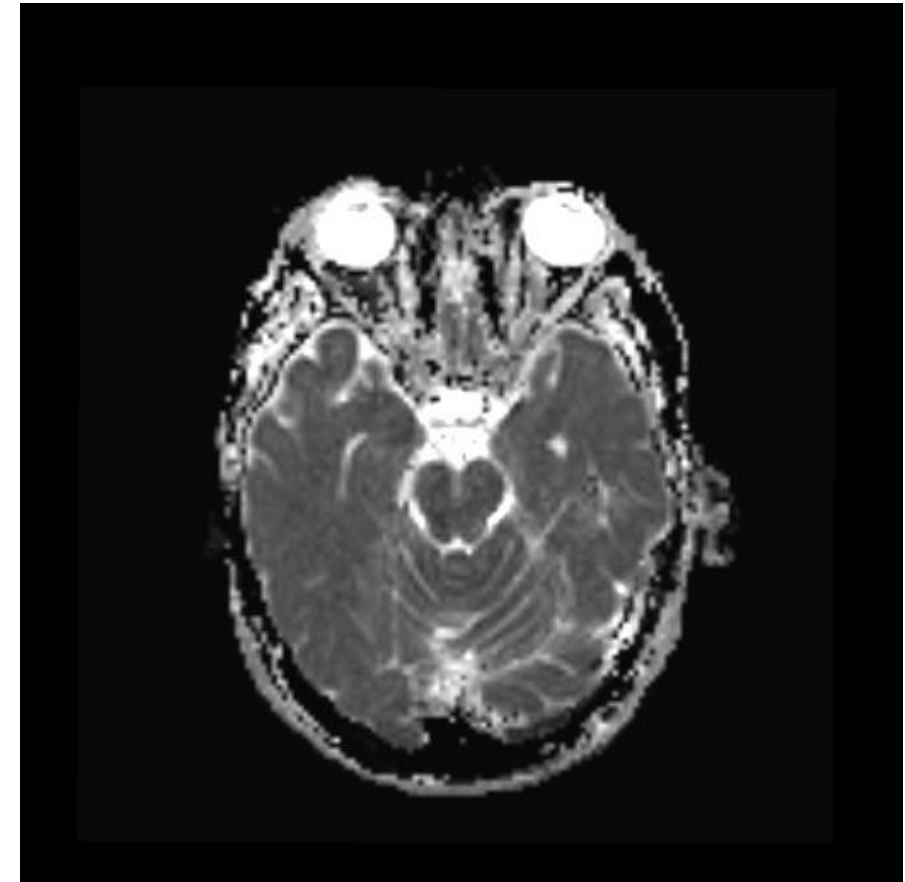


Development and exploratory testing of an analysis pipeline for ADC-derived radiomics biomarkers in brain tumor patients

BSc Semester Project

DWI in outcome prediction

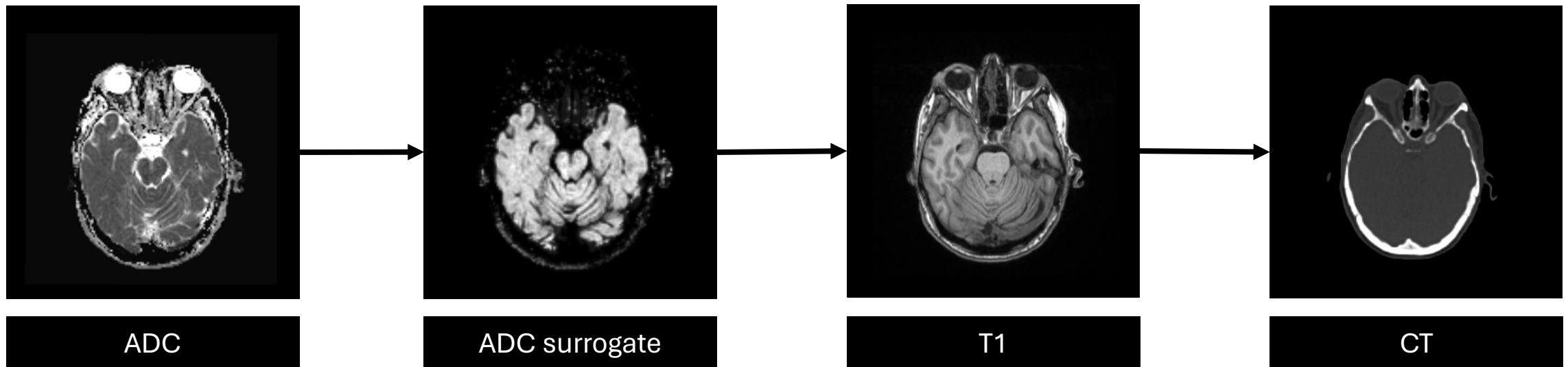
- Tumor characteristics that may be correlated with treatment outcome
- Diffusion weighted imaging (DWI) to investigate tumor characteristics (cellularity, intratumoral heterogeneity)
- Signal loss approximated as : $S(b)/S_0 = \exp(-b \cdot ADC)$
- ADC maps summarize diffusion information
- Require dedicated processing



Registration of ADC maps to planning CT images

Purpose

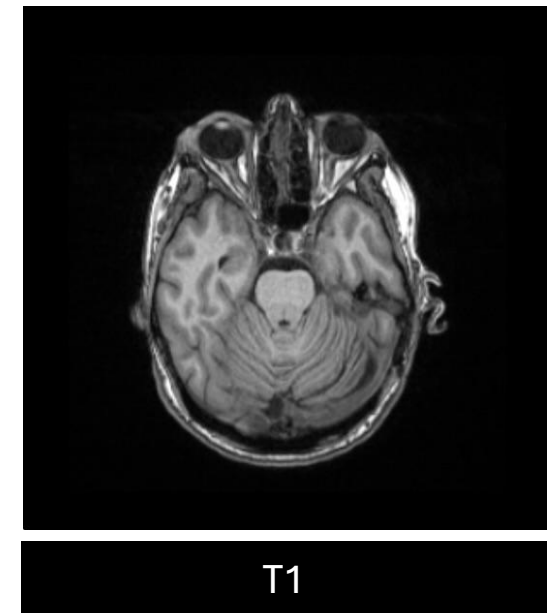
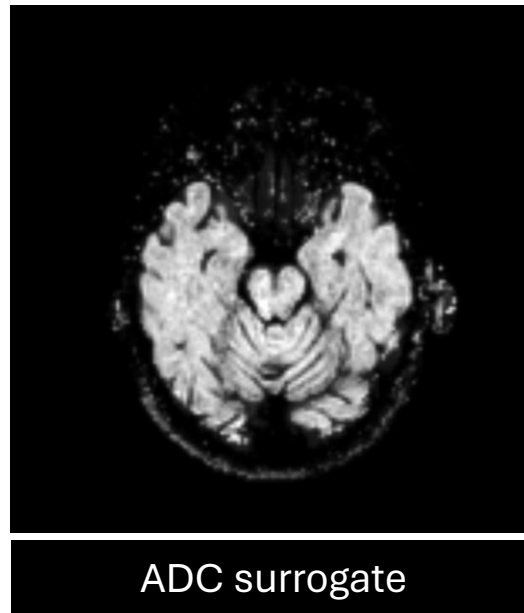
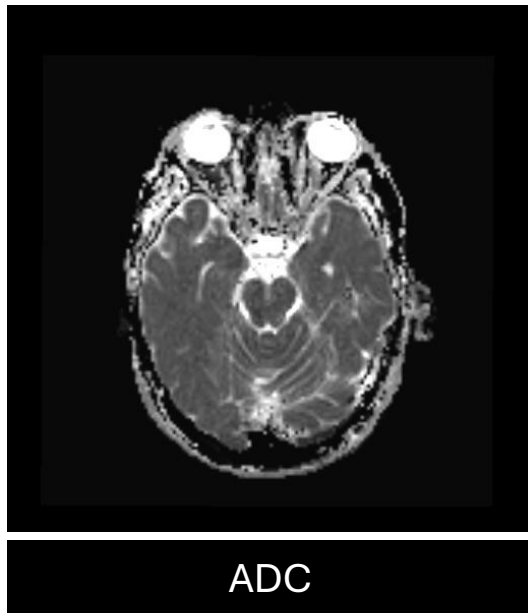
- Pooling of information on planning CT
- Allows for use of clinical information such as ROI contours and dose distributions



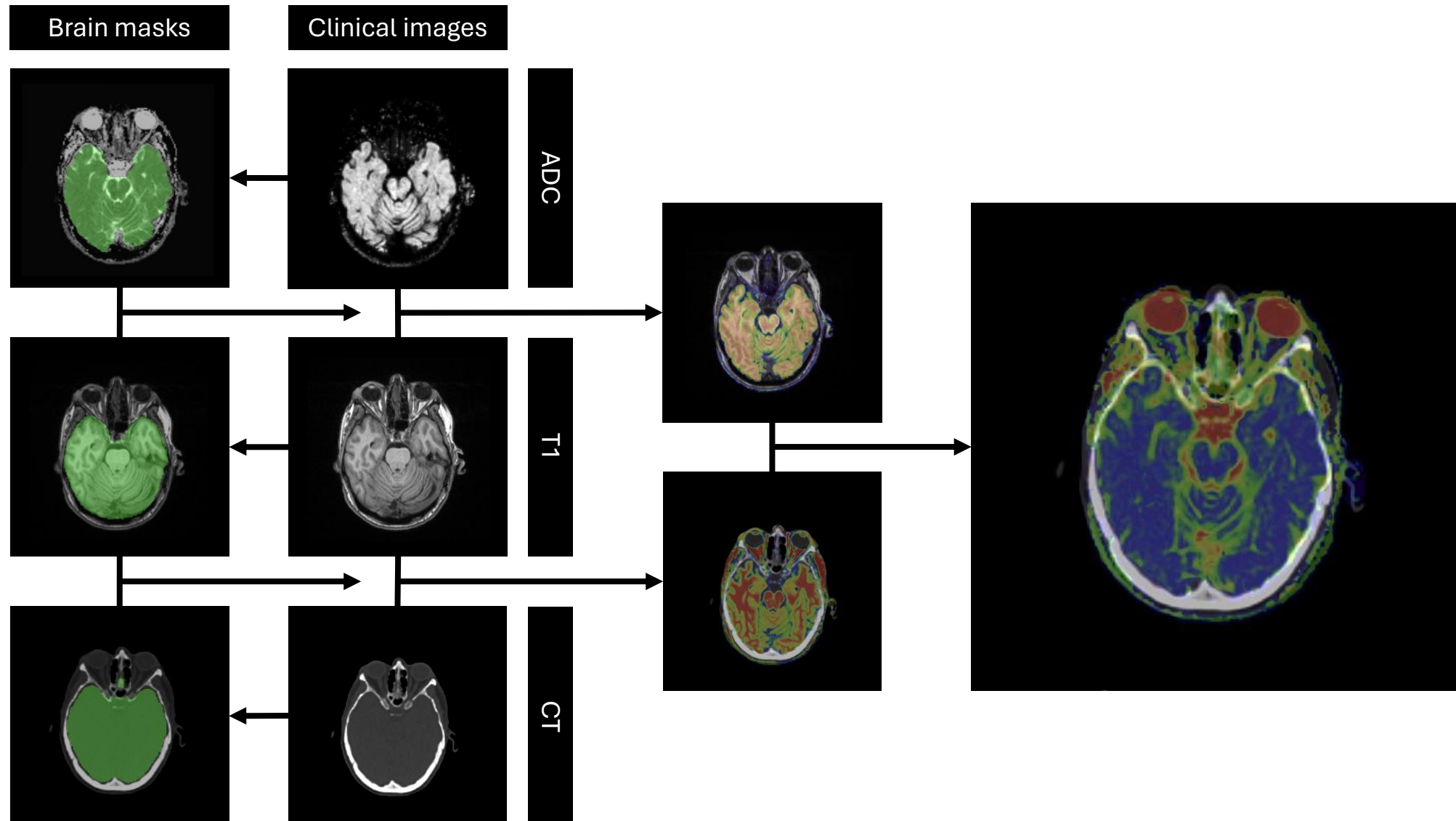
Anatomical-alike surrogate for ADC

Generation process

- Set values outside normal brain ADC value range $(0.3 - 1.5) \times 10^{-3} \text{mm}^2/\text{s}$ to interval edges
- Normalize range to interval $[0,1]$
- Invert non-zero values
- Gaussian smoothing



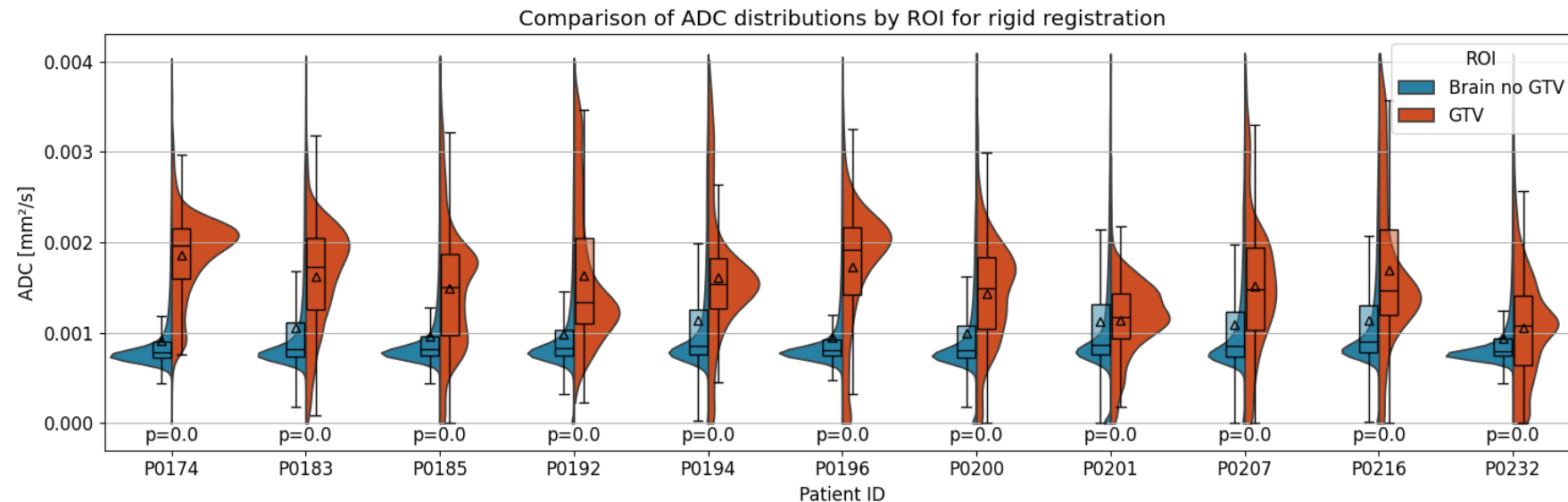
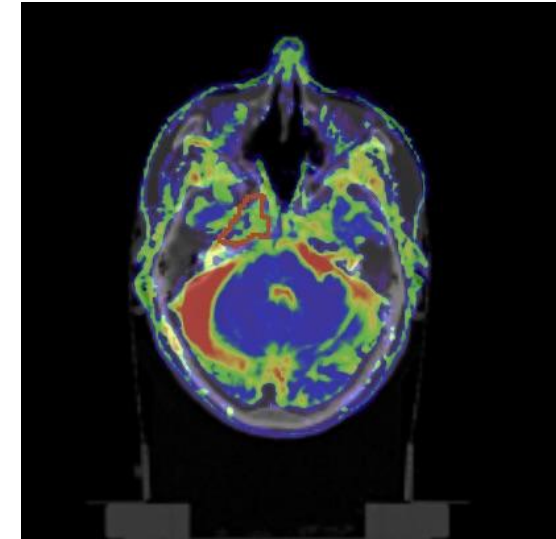
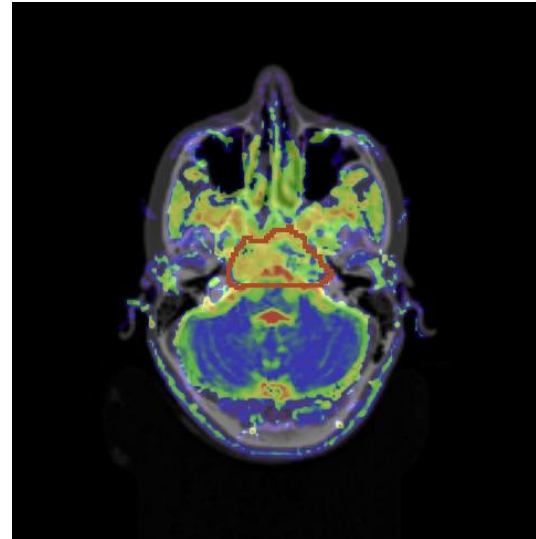
Registration Framework



Regional Comparison of ADC

Patient cohort

- 11 skull base tumor patients (chordoma & chondrosarcoma)
- Treated at PSI with IMPT
- 3 experienced local treatment failure

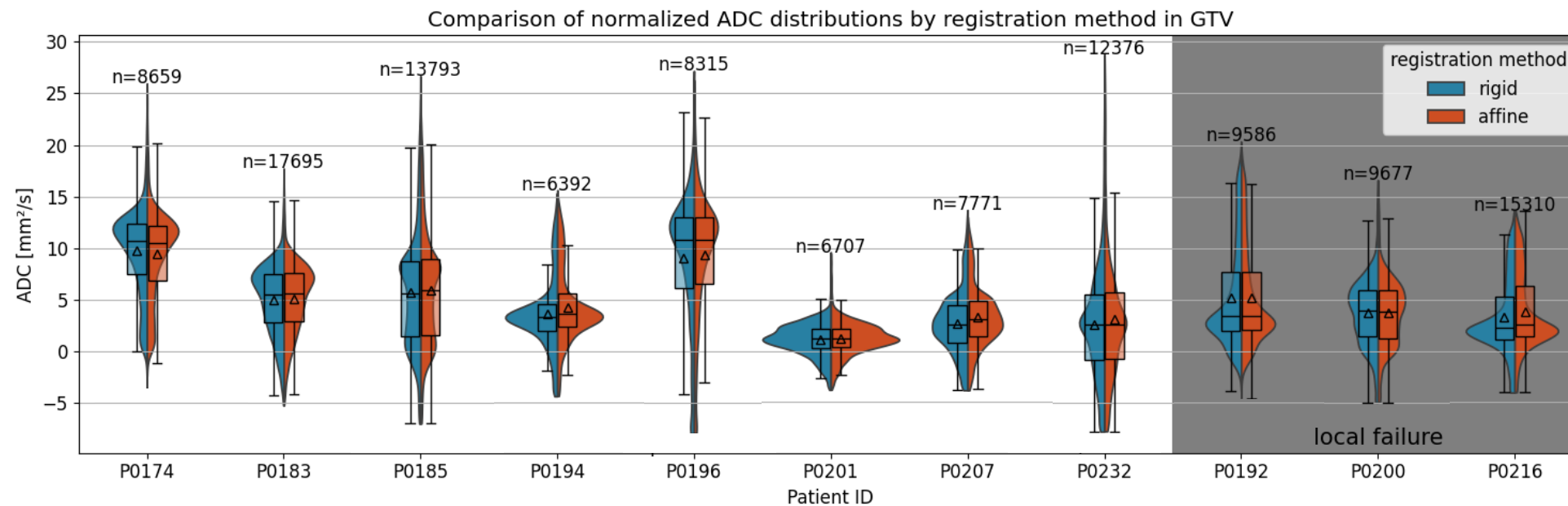


Comparison of ADC by Treatment Outcome

Inter-patient comparison

- ADC map harmonization
- No immediate differences between outcome groups

	Median ADC		Skewness ADC		Kurtosis ADC	
	no LF	LF	no LF	LF	no LF	LF
Median	4.4393	3.4068	0.062	0.6518	0.5388	0.1311



Correlation of Radiomics Features with Treatment Outcome

Feature extraction

- Pyradiomics
- 53 features per patient

Results

- ADC heterogeneity seems to be positively correlated with local treatment failure
- No correlation reached statistical significance

0.52 [0.0, 0.85]	0.1454	- firstorder Entropy
0.52 [-0.1, 0.86]	0.135	- gldm GrayLevelVariance
0.45 [-0.15, 0.85]	0.1882	- firstorder_Skewness
0.45 [-0.13, 0.85]	0.195	- ngtdm Contrast
0.45 [-0.18, 0.85]	0.1896	- glszm_SmallAreaHighGrayLevelEmphasis
0.39 [-0.2, 0.85]	0.2776	- glszm_HighGrayLevelZoneEmphasis
0.39 [-0.12, 0.84]	0.2806	- gldm_SmallDependenceHighGrayLevelEmphasis
0.39 [-0.41, 0.86]	0.2816	- glszm_GrayLevelVariance
0.39 [-0.2, 0.8]	0.2826	- glszm_SmallAreaEmphasis
0.39 [-0.2, 0.8]	0.2826	- glszm_SizeZoneNonUniformityNormalized
0.32 [-0.32, 0.79]	0.385	- gldm_LargeDependenceLowGrayLevelEmphasis
0.32 [-0.33, 0.79]	0.3904	- glszm_LargeAreaLowGrayLevelEmphasis
0.32 [-0.53, 0.85]	0.371	- gldm_DependenceEntropy
0.32 [-0.53, 0.86]	0.3766	- glszm_ZoneEntropy
-0.32 [-0.84, 0.2]	0.3688	- firstorder Uniformity
-0.32 [-0.84, 0.41]	0.3722	- ngtdm_Complexity
-0.39 [-0.86, 0.41]	0.2762	- glszm_GrayLevelNonUniformityNormalized
-0.39 [-0.8, 0.2]	0.2778	- gldm_SmallDependenceLowGrayLevelEmphasis
-0.39 [-0.79, 0.2]	0.2758	- firstorder Kurtosis
-0.39 [-0.8, 0.2]	0.2836	- glszm_LowGrayLevelZoneEmphasis
-0.39 [-0.8, 0.2]	0.2716	- glszm_SmallAreaLowGrayLevelEmphasis

spearman_rho

p_val

Analysis Pipeline

- Registration of ADC maps to planning CT images
- Extraction of ADC distributions from ROI
- Harmonization of ADC maps
- Extraction of radiomics features

Limitations

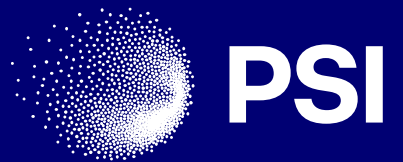
- Registration framework only validated via comparison of results from rigid and affine registration

Exploratory testing

- Elevated ADC in tumor volume
- Multiple features quantifying ADC heterogeneity show moderate correlation with treatment outcome

Limitations

- Small and imbalanced patient cohort
- Correlations did not reach statistical significance
- Reproducibility of Radiomics findings



Thank you for listening!

C. Tresch
PSI, 7. July 2025